

THE CUSTOMER'S PAIN

You see a charge you never made. Then you typically wait 37 hours to hear back.

A customer who reports a charge they do not recognize waits a **median of 37 hours** for the bank's first response. One in ten waits **58 hours or more**. Until then, the charge sits on their account with no answer.

37 h

median wait for a first response to "Cargo no reconocido" (unrecognized charge)

MEASURED

p90 58 h

n = 7,567 of 12,297

Why disputes: a dispute can be checked in code against the customer's own ledger and decided by an explicit policy.

DESIGN ARGUMENT

Source: docs/analysis/demand-report.md (finding 3: median 37.0 h, p90 58.0 h over the 7,567 of 12,297 complaints with a recorded first response). Measured on the challenge's synthetic LATAM Bank dataset, not a live bank.

A dispute agent that knows when to act, when to ask, and when to step aside.

1 Resolves, with proof

RESUELTO

✓ **Identidad verificada**
Sesión de demostración activa

✓ **Transacción localizada**
Cargo identificado: Uber · COP 38.500 · 14 de jun de 2026

✓ **Política evaluada**
Resolución automática aplicada

✓ Referencia REF-7B47679128

The record backs the claim, so a provisional credit lands in seconds, in the same chat.

2 Asks when it is unclear

Veo que tiene 2 cargos que coinciden con lo que contó. ¿Cuál es el que no reconoce? Toque directamente en la lista que aparece debajo, o seleccione "No está en la lista" si no lo ve allí.

Taxi Seguro
COP 27.000
15 de jun de 2026 · Transporte

Taxi Seguro
COP 27.000
15 de jun de 2026 · Transporte

No está en la lista

Escriba su mensaje...

Two matching charges: the customer picks from their own. It never guesses.

3 Protects, hands off

ACCIONES REALIZADAS

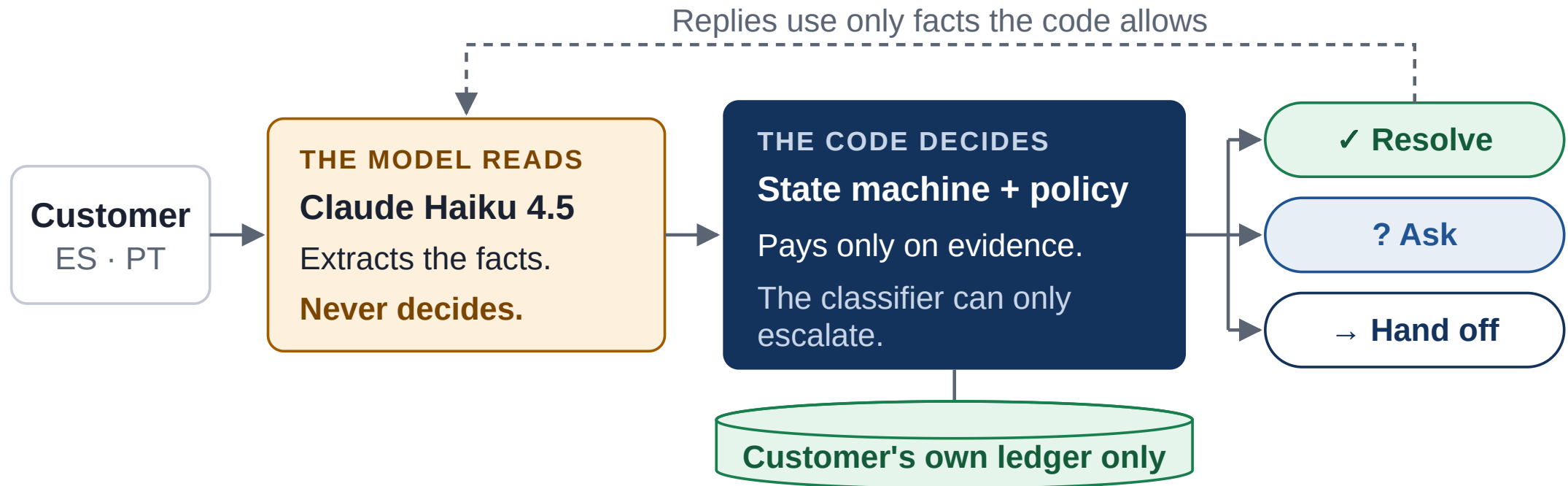
El cliente nombró el comercio y es su único cargo que coincide. El cargo no cumple las condiciones de auto-resolución.

Tarjeta bloqueada preventivamente por el agente (simulado, sin movimiento de dinero).

Likely fraud: the card is blocked first, then a person gets a complete file of verified facts.

Spanish and Portuguese Credits and card blocks are simulated

The model reads. The code decides.



The LLM only extracts and phrases
No refund comes from the model.

Policy and permissions live in code
No lookup can reach another customer's ledger.

Verified facts only
A persuasive claim the record contradicts still goes to a person.

Design argument, enforced in code: an exhaustive sweep over every reason and all 128 combinations of the other conditions shows the classifier never causes a credit (tests/test_policy_not_overridden.py) · 1,121 automated tests · docs/architecture-decisions.md dispute-agent AD-3, AD-5, AD-13 to AD-15.

Honest models: we ship only what the numbers back.

ETL with contracts

CONTRACTS

9 tables, one schema contract each

QUALITY CHECKS

Row counts, duplicate keys, null rates, foreign keys, and a lineage manifest

FOREIGN KEYS MEASURED

✓ **0 orphaned rows**

Why a subset

FOR ANALYSIS MEASURED

130,690 of 5M transactions (May 18 to Jun 17, 2026); **67,095** complaints in full

LIVE APP

1 customer, **16 charges** (6 real, 10 synthetic)

WHY DESIGN ARGUMENT

No S3 at runtime (privacy), reproducible, free-plan deploy; a dispute only needs the recent ledger

Fraud model vs the bank

DATA MEASURED

2,951,642 transactions over 2 years, split by date

TEST PR-AUC MEASURED

Bank	0.720
Ours	0.707

SHIPPED INSTEAD MEASURED

Gate **fraud score > 30**, chosen by cost (some inputs assumed). The model only informs the advisor.

Our model lost to the bank's score, so it never decides: the rule the numbers back does.

Sources: etl/schema_contract.py, data/lineage_manifest.json, data/extraction_manifest.json; docs/ml/fraud-model.md, AD-15 (on test, the gate's 48 escalations are all fraud). The priority classifier shows no measured lift (p = 0.45) and can only escalate.

48 held-out conversations on the real model. Zero unsafe outcomes.

0 / 48

unsafe, in each of 3 runs

MEASURED

18 / 18 resolvable cases resolved

26 / 26 card-block calls right

3.0 s median reply (p95 7.7 s)

\$0.004 per case

SAME 48 CASES	MEASURED	CORRECT OF 48	UNSAFE OF 48	UNNEEDED TRANSFERS OF 18
Our hybrid				
Claude Haiku 4.5, 3 runs		43	0	0
Regex extractor				
extraction swapped for rules, 1 run		43	0	0
Always escalate				
safety anchor, from labels		n/a	0	18

Escalating everything is safe but helps no one. The 5 misses moved no money.
Mocked-model suite: without the evidence check, 6 unsafe credits. SIMULATED

Read with its limits: one customer's account, messages written by a model that never saw the policy, labels fixed before the first run. 0 of 48 bounds the unsafe rate below about 7% (95%), not at zero.

ROUTE TO PRODUCTION

From demo to the bank's front line.

STEP 1

Real identity

The bank's identity verification replaces the demo login.

STEP 2

Managed database

Replicated storage across several instances, not local SQLite.

STEP 3

Monitoring

The structured, correlated event log feeds alerting.

STEP 4

Real traffic

Thresholds and the live model re-validated on real customer traffic.

LIVE DEMO

factored-hackaton-latest.onrender.com

CODE

github.com/jAgusGelos/factored-hackathon-2026-jagusgelos

An answer in seconds, not 37 hours. Every refund backed by evidence. Every hard case in a person's hands, with the file already complete.